Ankit Kulshrestha

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Education

University of Wisconsin-Madison — Madison, WI
B.S. in Computer Science and B.S. in Data Science — *Sep. 2020 – May 2024*
GPA: 3.5

Experience

Software Engineer — July 2024 – August 2026
Holos, Inc. — *Madison, WI*

- Designed and developed interactive educational applications using C#, Unity, and SwiftUI, following agile workflows across the full development lifecycle
- Integrated Scale AI's Donovan platform via REST API to automate AI-generated educational course content, building a data pipeline that structured LLM outputs and cut course production time by 85%
- Built backend data management systems in Unity and C# to parse, store, and manage structured application data using JSON-based local persistence, reducing load times by 45%
- Wrote and maintained 150 unit and integration tests and performed automated and manual cross-platform testing to ensure stable application builds

Software Engineer — June 2023 – May 2024
Zamanian Lab — *Madison, WI*

- Developed image and video processing pipelines in Python using OpenCV for high-throughput parasitic disease drug screening across 24–96 well plates and videos of up to 500 frames, supporting research for industry partner Zoetis
- Refactored the architecture of wrmXpress, a computer vision library, improving processing efficiency by 30% for multi-modal image and video data including motility analysis via optical flow and worm segmentation
- Collaborated with researchers to translate experimental biology requirements into software specifications

Software Engineering Intern — June 2022 – Aug. 2022
Cognex Corporation — *Milwaukee, WI*

- Collaborated with an Agile team of nine to develop features and fix bugs using C# and XAML in a machine vision system application
- Created a Windows Presentation Foundation control using XAML to ensure that developers would be able to create warning, error, and informational messages easily

Projects

RunItBack - Smash Bros Replay Analyzer | *Python, PyTorch, OpenCV* — Dec. 2025 – Present

- Developed ML-powered video analysis tool that detects game boundaries, classifies 7 positional scene states, and tracks key events in fighting game replays
- Fine-tuned ConvNeXt-Tiny CNN via transfer learning on custom-labeled video replay data, achieving 92% scene classification accuracy; exported model to ONNX Runtime for inference optimization
- Engineered a multi-stage computer vision pipeline combining PaddleOCR with CLAHE preprocessing for damage tracking, template matching for character identification, and Gemini Vision API integration with response caching
- Built a PyQt5 video labeling application and coordinated a 70-member Discord community to crowdsource training data annotation, designing the data collection workflow for non-technical contributors

GenModel - AI-Powered 3D Model Generation | *Swift, RealityKit, REST APIs* — Dec. 2025 – Jan. 2026

- Integrated 2 ML models (OpenAI Shap-E, TRELLIS) via Replicate's REST API, supporting both text-to-3D and image-to-3D generation pipelines with $0.01–0.05 per request cost optimization
- Built a thread-safe API client using Swift's actor concurrency model, handling async polling with exponential backoff (1s–5s intervals, 300-attempt timeout) for long-running ML inference jobs

Technical Skills

Languages: Python, C#, C, Swift, Java, SQL, R, HTML, CSS, JavaScript
Developer Tools: Git, Docker, Unity, AWS, Xcode, pytest, VS Code
Libraries: PyTorch, OpenCV, pandas, NumPy, Flask, React, SQLAlchemy, Matplotlib, PaddleOCR
Other: Generative AI/LLMs, Prompt Engineering